

# AP CSP | Lesson 19: Procedures & Libraries

## Homework | Multiple-choice and written practice

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Q5: select exactly TWO. Procedure definitions and library contracts are supplied in the question. RETURN sends a value to the caller; DISPLAY produces output. Put braces on separate lines.

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1. How many times does FOR EACH run its body for an empty list? [1 mark]

A. 0  
B. 1  
C. The previous list length  
D. It must access index 1

Answer: \_\_\_\_\_

2. A supplied procedure toHours(m) returns  $m / 60$ . What does toHours(150) return? [1 mark]

A. 150  
B. 2  
C. 2.5  
D. 90

Answer: \_\_\_\_\_

3. For PROCEDURE combine(left, right), the call is combine(4, 9). Which statement is correct? [1 mark]

A. 4 and 9 are parameter names.  
B. left receives 4 and right receives 9.  
C. left receives 9 and right receives 4.  
D. Both parameters must be named combine.

Answer: \_\_\_\_\_

4. A procedure only DISPLAYs a number and has no RETURN. What can its caller safely assume? [1 mark]

A. The displayed number is its returned numeric value.  
B. The program must stop completely.  
C. No output is produced.  
D. Output is produced, but no numeric return is promised.

Answer: \_\_\_\_\_

5. Which TWO statements are true? Select exactly TWO answers. [1 mark]

A. A parameter can let one procedure work with varied inputs.  
B. Calling a library removes the need to read its documentation.  
C. A procedure definition executes its body immediately.  
D. Documentation can specify parameters and returned values.

Answer: \_\_\_\_\_

6. Inside a procedure, RETURN(6) is executed before DISPLAY(99). What happens? [1 mark]

A. 99 is displayed before returning 6.  
B. 6 is returned and that later DISPLAY is skipped.  
C. The procedure returns 99.  
D. Every remaining caller statement is skipped.

Answer: \_\_\_\_\_

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### Written responses | Questions 7-9

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7. [4 marks]

toHours(m) returns  $m/60$ ; reachedGoal(m,g) returns  $m \geq g$ . Give (a) toHours(45), (b) toHours(120), (c) reachedGoal(30, 30), and (d) reachedGoal(30, 45).

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8. [4 marks]

Write a procedure toHours(minutes) that returns minutes/60. Then write one call that assigns its result for 90 minutes to hours. Do not print inside the procedure.

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9. [4 marks]

Write reachedGoal(minutes, goal) to return whether minutes  $\geq$  goal. Then assign reachedGoal(20, 30) to met. Use both parameters; do not hard-code a fixed target.

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### Written responses | Questions 10-12

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**10.** [4 marks]

A procedure demo executes RETURN(8) and then contains DISPLAY(99). A caller evaluates result  $\leftarrow$  demo() + 2. Give (a) the returned value, (b) whether 99 is displayed, (c) why the later statement is skipped, and (d) result.

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**11.** [4 marks]

Fictional DurationKit provides seconds(m): m is a nonnegative number of minutes; it returns numeric seconds and produces no display output. State (a) its input unit, (b) returned unit, (c) seconds(2.5), and (d) whether a separate DISPLAY is needed to show that result.

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**12.** [4 marks]

A program copies the same goal comparison into three features. (a) Name a reusable helper. (b) Propose its two parameters. (c) Explain one maintenance benefit. (d) State the condition under which changing its internals need not change its callers.

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